

Water Source HR Outdoor Unit

| Component                             | Abbreviation |           | Cooling Main Mode                                  |                                                                                                                |
|---------------------------------------|--------------|-----------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Compressor                            | Comp         | Operation | Target low pressure control (default: 101.5 psig)  | Primary function, Maintain Average Evaporator Temperature (Average temperature of Eva. In Sensors)             |
|                                       |              | Range     | 14~120Hz                                           | Low pressure range 100psig - 128 psig. Evaporator temperature range 45°F -48°F                                 |
| High side pressure                    | HP           | Operation | Head pressure is effected by the water temperature | Is there a variable flow control valve installed? This will effect head pressure.                              |
|                                       |              | Range     | 299 psig - 469 psig (water temperature 68°F-104°F) | At 228psig, low limit control is activated at 512psig high limit control is activated                          |
| Main EEV                              | E_M          | Operation | Cooling & Heating capacity balance control         |                                                                                                                |
|                                       |              | Range     | 300 - 2000 Steps                                   |                                                                                                                |
| EVI EEV                               | E_EVI        | Operation | *Subcooling control (default: SC=36°F)             | SC = High Side Saturation Temp. - Liquid Line Temp.                                                            |
|                                       |              | Range     | 0-480step                                          |                                                                                                                |
| 4 Way Valve                           | V_4W         | Operation | Off                                                |                                                                                                                |
| EVI By-Pass Solenoid Valve            | V_EB         | Operation | On (Open): *T_W > 104°F or 50Hz↓                   | Activates Sub-Cooling Control                                                                                  |
|                                       |              |           | Off (Close): T_W < 100.4°F↓ and 65 Hz↑             | Discharge Temperature Protection, Control by unit                                                              |
| EVI Solenoid Valve                    | V_EVI        | Operation | On (Open)                                          |                                                                                                                |
| Hot Gas By-Pass                       | V_HG         | Operation | Off (Closed)                                       |                                                                                                                |
| Accumulator Oil Return Solenoid Valve | V_AR         | Operation | On (Open when compressor is running)               | <b>Off</b> during start up if DSH is < 18°F. <b>Off</b> for 2 min. if DSH is < 18°F <b>On</b> when DCSH > 27°F |
| Hot Gas By-Pass 2 (HR Only)           | V_HG2        | Operation | Off Closed                                         | Opens after 20 minutes of continuous run time in cooling mode                                                  |
| Main EEV Solenoid Valve (HR Only)     | V_ME         | Operation | On (Open)                                          | Energized in cooling only                                                                                      |
| Main Cooling Solenoid                 | VMC          | Operation | On (open)                                          | Directs hot gas to MCU for zones calling for heat                                                              |
| Liquid Pipe Thermistor                | T_L          | Value     | Used to calculate sub-cooling                      | Sub-Cooling range is 9°F-36°F                                                                                  |
| HP - HR Ball Valve (3 phase only)     | BV_HP        | Operation | Open                                               | Valve must be open for heat pump applications                                                                  |

Mode Control Unit (MCU)

| Component                  | Abbreviation |           | Cooling Mode                        | Comments                                                             |
|----------------------------|--------------|-----------|-------------------------------------|----------------------------------------------------------------------|
| Port Heating Solenoid      | Heat         | Operation | Off (Closed) ,On (Open)             | Energized for units calling for heating                              |
| Port Cooling Solenoid      | Cool         | Operation | Off (Closed) ,On (Open)             | Energized for units calling for cooling                              |
| Port EEV                   | EEV          | Operation | Open 480 Steps                      | Open 480 step when a zone calls for cooling or during a mode change. |
|                            |              | Range     | 0 - 480 Steps                       | Closed 0 Steps when a zone is in heating                             |
| Sub-Cooling EEV            | SC_EEV       | Operation | Closed 0 Steps                      | Provide subcooling for cooling zones                                 |
|                            |              | Range     | 480 Steps                           | Maintains 5.4 degrees of superheat between pipe in and out temps.    |
| Sub-Cooler In Temperature  |              | Operation |                                     |                                                                      |
| Sub-Cooler Out Temperature |              | Operation | Close to Suction Header Temperature |                                                                      |

Indoor Units

| Component              | Abbreviation |           | Cooling Mode                                                | Comments                                                             |
|------------------------|--------------|-----------|-------------------------------------------------------------|----------------------------------------------------------------------|
| Room Temperature       | Room Temp.   | Operation | 65~86°F                                                     |                                                                      |
| Liquid Pipe Thermistor | EVA In 1     | Operation | Evaporator Saturation Temperature. Range 45-48°F            | Used to calculate Evaporator Superheat                               |
| Gas Pipe Thermistor    | EVA Out 1    | Operation | Measures Suction Pipe Temperature                           | Used to calculate Evaporator Superheat                               |
| EEV                    | EEV 1        | Operation | Modulates to maintain Superheat of 3.6 degrees F            | Superheat = Pipe out Temp - Pipe in Temp. (Typical Range 0-7)        |
|                        |              | Range     | 0 - 2000 Steps (0 Steps when zone is satisfied)             | Typical range between 250 and 1500 Steps                             |
| IDU Fan                | Fan Speed    | Range     | Low , Medium, High, Auto                                    |                                                                      |
|                        |              |           | Low Speed:  Ts - Ti  ≤ 1.8°F                                | TS - Set Temperature                                                 |
|                        |              |           | Medium Speed: 1.8°F <  Ts - Ti  ≤ 5.4°F                     | Ti Indoor Temperature                                                |
|                        |              |           | High Speed 5.4°F <  Ts - Ti                                 |                                                                      |
| Component              | Abbreviation |           | Heating Mode                                                | Comments                                                             |
| Room Temperature       | Room Temp.   | Operation | 61~86°F                                                     |                                                                      |
| Liquid Pipe Thermistor | EVA In 1     | Operation | Liquid Pipe Temperature for sub-cooling Control             | Used to calculate Evaporator Sub-Cooling                             |
| Gas Pipe Thermistor    | EVA Out 1    | Operation | Hot Gas Pipe Temperature should be close to discharge temp. | Used to evaluate the temperature of the hot gas coming into the coil |
| EEV                    | EEV 1        | Operation | Modulates to maintain 9 degrees of sub-cooling              | Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.       |
|                        |              | Range     | 500- 1500 Steps (80 - 120 Steps when zone is satisfied)     | Typical range between 500 and 1500 Steps                             |
| IDU Fan                | Fan Speed    | Range     | Low , Medium, High, Auto                                    |                                                                      |
|                        |              |           | Low Speed:  Ts - Ti  ≤ 1.8°F                                | TS - Set Temperature                                                 |
|                        |              |           | Medium Speed: 1.8°F <  Ts - Ti  ≤ 5.4°F                     | Ti Indoor Temperature                                                |
|                        |              |           | High Speed 5.4°F <  Ts - Ti                                 |                                                                      |